MIL-T-52964(ME)
23 July 1979

MILITARY SPECIFICATION

TOPOGRAPHIC SUPPORT SYSTEM (TSS):

OPERATION, STORAGE AND RETRIEVAL, DISTRIBUTION,

DRAFTING SUPPORT, COMPILATION, SURVEY, RECTIFIER I,

RECTIFIER II, MOSAICKING/DRAFTING, COLLECTION,

ANALYSIS, INFORMATION AND SYNTHESIS SECTIONS,

SEMITRAILER MOUNTED, GENERAL SPECIFICATION FOR

This specification is approved for use by the Mobility Equipment Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the general requirements for 13 Topographic Support System (TSS) sections. Each section consists of: a semitrailer (chassis); International Standardization Organization (ISO) container (van body) with air conditioning, heating, and utilities, installed and equipment and materials listed in the applicable set listings mounted therein.

1.2 Classification. The TSS shall be of the following types of sections, as specified (see 6.2):

1. Operation Section
2. Storage and Retrieval Section
3. Distribution Section
4. Drafting Support Section
5. Compilation Section
6. Survey Section

FSC 6675

Beneficial comments (recommendations, additions, deletions) and any other pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, Virginia 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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7. Rectifier I Section
8. Rectifier II Section
9. Mosaicking/Drafting Section
10. Collection Section
11. Analysis Section
12. Information Section
13. Synthesis Section

2. APPLICABLE DOCUMENT

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

| | |
|---|---|
| PPP-B-601 | - Boxes, Wood, Cleated-Plywood. |
| PPP-B-1055 | - Barrier Material, Waterproofed, Flexible. |
| PPP-T-75 | - Tape, Preservation and Sealing. |

MILITARY

| | |
|---|---|
| MIL-P-116 | - Preservation-Packaging, Methods of. |
| MIL-B-121 | - Barrier Material, Greaseproofed, Waterproofed, Flexible. |
| MIL-P-514 | - Plates, Identification, Instruction and Marking, Blank. |

STANDARDS

MILITARY

| | |
|---|---|
| MIL-STD-129 | - Marking for Shipment and Storage. |
| MIL-STD-130 | - Identification Marking of US Military Property. |
| MIL-STD-281 | - Trailers and Trailer Dollies; Preservation and Packaging of. |
| MIL-STD-461 | - Electromagnetic Interference Characteristics Requirements for Equipment. |
| MIL-STD-810 | - Environmental Test Methods. |
| MIL-STD-1186 | - Cushioning, Anchoring, Bracing, Blocking and Waterproofing; with Appropriate Test Methods. |



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MIL-STD-1188 - Commercial Packaging of Supplies and Equipment.

MIL-STD-1472 - Human Engineering Design Criteria for Military Systems, Equipment and Facilities.

See Supplement 1 for list of associated specification sheets.

(Copies of specifications, standards, and set listings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

DEPARTMENT OF TRANSPORTATION (DOT)

49CFR571 - DOT Federal Motor Vehicle Safety Standards (FMVSS).
49CFR325, 390, & 393 - DOT Federal Motor Carrier Safety Regulations (FMCSR).

(Applications for copies should be addressed to a local DOT office or the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification.

(Application for copies should be addressed to the Amercian Trucking Associations, Inc., ATTN: Traffic Department, 1616 P Street, N.W., Washington, D.C. 20036.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE Handbook.

(Applications for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

STATE OF CALIFORNIA DEPARTMENT OF MOTOR VEHICLES

California Vehicle Code.



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(Application for copies should be addressed to the State of California, Department of Motor Vehicles, 2570 24th Street, Sacramento, CA 95804.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, ATTN: Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

TRUCK-TRAILER MANUFACTURERS ASSOCIATION (TTMA)

TTMA Recommended Practices.

(Application for copies should be addressed to the Truck-Trailer Manufacturers Association, Inc., 2430 Pennsylvania Avenue, N.W., Washington, DC 20037.)

## 3. REQUIREMENTS

3.1 Description. The sections shall be as specified in the applicable specification sheets and as specified herein.

3.1.1 Specification sheets. The individual section requirements shall be detailed in the applicable specification sheets.

3.2 First article (preproduction model). The contractor shall furnish one of each type section for examination and testing within the time frame specified (see 6.2) to prove prior to starting production that his production methods will produce sections that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4; and, unless otherwise specified herein, all examinations and tests shall be conducted by the contractor subject to surveillance and approval by the Government (see 6.3). When specified (see 6.2), the Government will conduct any or all of the preproduction examinations and tests, as specified (see 6.2).

3.3 Initial production. When specified (see 6.2), the contractor shall furnish to the Government one or more of each type sections for inspection as specified in 4.4.

3.4 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.4).



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3.5 Fabricating and mounting of equipment. The contractor shall furnish all equipment shown in the sections. The contractor shall mount and install all equipment in the sections. Any holes cut in the van body shall be sealed and made light and waterproof. Provisions shall be made in all sections to secure all items (chairs, equipment cases, map files, accessary cases, and magnifier lamps). Also to secure (keep from opening during transport) all filing cabinets: legal, map and plan, and shelves. This shall be accomplished by bars, clasps or other government approved devices. Provide recessed folding steps to gain access to the van roof and exhaust fan cover. The mountings shall not affect the proper functioning of the equipment when the section is stopped, leveled, and set up for normal field usage, and shall withstand the shock and vibration encountered in travel and transport by rail, sea, and air.

3.6 Electrical. Receptacle locations shall be located to permit their access with the installation of equipment in accordance with the specification sheets. Overhead light fixtures shall be sized and located to provide minimun interference to operating personnel in the section and to their access to installed equipment. The section shall have the required voltage and phase distribution (split, three or single) for the electrical operation of the components installed in the sections. The electrical equipment shall be effectively guarded and grounded to protect all personnel and objects from electrical shock hazard.

3.7 Human Engineering. All components (except the designated manufacturer make and models) and the design effort (within the scope of the Contractor) shall meet the requirements of MIL-STD-1472:

General Requirements
Work Space
Storage Space
Standing Operations
Work Surface Width
Work Surface Height
Illuminace
Design for Maintainability - General Criteria
Adjustment Controls
Lubrication
Access Opening and Covers
Vans

3.8 Interchangeability. Common components shall be used to the maximum extent practical in the sections to minimize the number of line-item repair parts.



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3.9 Safety. All belts, gears, shafts, chains, and other reciprocating or rotating parts of equipment shall be guarded when such parts are exposed to contract by personnel or otherwise create a hazard. Electrical equipment shall be guarded and grounded to protect all persons and objects from electrical shock hazard. Nonfunctional sharp edges, projecting points, and excessive length of fastening devices shall be avoided. Protect (rubber corners) all wall cabinet lower corners that are exposed to personnel traffic A covering (to reduce static electricity shock) shall be installed and secure to the floor. The elastomer covering shall be green in color and the area is approximately 240 square feet. It is required in all the sections except the Storage and Retrieval and Distribution.

3.10 Maintenance. All major assemblies and installed attachments shall be accessible for maintenance, repair, and replacement without the removal of other major assemblies and installed attachments not normally removed. Drain outlets shall be located for accessibility. Provisions shall be made to permit the use of receptacles for collecting drainage.

3.11 Performance.

3.11.1 Low and high temperature storage. The sections shall not be damaged (see 6.9), by storage in any ambient temperature from minus 34° C (-30° F) to plus 71° C (+160° F).

3.11.2 Mobility. When loaded with the tools and equipment specified herein, the sections shall withstand driving on paved roads, unimproved roads, and cross-country trails. During or following driving, there shall be no damage to (see 6.9), or displacement of any component, accessory, equipment, part, or tool installed in or on the sections and no evidence of defective welded, bolted, or riveted connections.

3.11.2.1 Highway travel. The sections shall travel over smooth, hard-surfaced, primary highways at sustained speeds between 45 and 55 mph.

3.11.2.2 Secondary road travel. The sections shall travel over gravel-surfaced secondary roads at sustained speeds between 30 and 40 mph.

3.11.2.3 Off-road travel. The sections shall travel over unimproved roads, open fields, rolling hills, and rough off-road terrain at speeds between 10 and 20 mph.

3.11.2.4 Chatter course. The sections shall travel, without damage (see 6.9), or permanent deformation, over a chatter course that will produce upward or downward accelerations of the magnitude of between 3 to 6 g's to a rear wheel center of the vehicle.



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3.11.2.5 Diagonal V-ditch course. The sections shall travel, without damage or permanent deformation, over V-ditches to produce not less than 60 reversal torsional frame stresses.

3.11.2.6 Mud course. The sections shall negotiate a mud course not less than 12 inches in depth, and shall withstand immobilization and towout operations.

3.11.2.7 Braking. The braking performance of the sections shall be in compliance with DOT MVSS, DOT FMCSR, and the California Vehicle Code.

3.11.2.8 Semitrailer mobility. ISO van bodies installed on container semitrailers shall, in addition, withstand pivot turns about the rear suspension, comply with DOT FMCSR off-tracking, provide clearance for jackknife turns not less than 90 degrees from the truck-tractor, and be compatible with commercial truck-tractors. The semitrailer brake system shall also comply with TTMA RP No. 44.

3.11.3 Rail impact test. When specified (see 6.2), the mounted section shall withstand tail humping without damage or permanent deformation to parts or components when tested as specified in 4.6.2.4.

3.11.4 Precipitation. When tested as specified in 4.6.2.5, the van bodies shall not leak water into the interior of the van body when exposed to rain ranging from a drizzle to a heavy rain of 1/4 gallon per minute per square foot of wetted surface impinging on the van body with a 30 mph wind. During and after the precipitation, the equipment shall operate.

3.11.5 Leveling. When tested as specified in 4.6.2.6, the van bodies shall have the capability of being leveled (within 15 minutes) on rough terrain, including side slopes up to 5 percent.

3.11.6 Blackout. When tested as specified in 4.6.2.7, the van bodies shall be blackout-proof to prevent leakage of lighting from inside under blackout (BO) conditions.

3.11.7 Demounting. When tested as specified in 4.6.2.8, the van bodies shall be demountable from the chassis.

3.11.8 Electromagnetic interference. When specified (see 6.2), the electromagnetic interference (EMI) emission characteristics of the van body's utility line electrical system and the contractor-installed electrically operated equipment shall conform to MIL-STD-461, class IIB.

3.11.9 Preparation for operation and movement. When tested as specified in 4.6.2.13, the section shall be setup for field operation within two (2) hours; then prepared for transport within two (2) hours. The number of



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personnel used for this test shall not exceed the number assigned to operat the section.

3.12 Identification.

3.12.1 Markings. The sections shall be identified in accordance with MIL-STD-130. The markings shall be applied to the sections on plates conforming to MIL-P-514, Type I, Style 1, Composition C, of Type I, Grade A, Class 1 material. Plates shall be attached by screws, bolts, or rivets in a location where the plates will be both visible and legible.

3.12.2 Shipping data plate. A shipping data plate conforming to MIL-P-514, Type III, Composition C, of Type I, Grade A, Class 1 material shall be provided. In addition, the plate shall show the silhoutte of the section in transport position, indicating the center of gravity and the location and capacity of the slinging and tiedown provisions. The plates shall be attached by screw, bolts, or rivets in a location where the plates will be both visible and legible.

3.13 Government-loaned property. When specified (see 6.2), the following property in the quantities indicated will be loaned by the Government (see 6.5).

| Description | Identification | Quantity |
|---|---|---|
| Tractor, 5 ton | NSN 2320-00-050-8984 | 1 |

3.14 Doors. The doors on the van bodies shall be as indicated in the figure to the applicable specification sheet.

3.15 Workmanship. All parts, components, and assemblies of the section shall be clean of harmful extraneous material, such as sand, dirt, spurs, scale, and flux. External surfaces shall be free of burrs, sharp edges, and sharp corners, so as not to be detrimental to safety.

3.15.1 Bolted connections. Boltholes shall be accurately formed and shall have burrs removed. Washers or lockwashers shall be provided where necessary. Matching thread areas securing bolts conforming to SAE J429 or capscrews shall be of sufficient strength to withstand the tensile strength of the bolt. All fasterners shall be correctly torqued and shall have full thread engagement. Bolts shall protrude not more than two full threads.

3.15.2 Riveted connections. Rivets shall fill the holes completely. The upset rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member, and shall be in accordance with SAE J492.



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## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications, standards, and set listings, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Initial production inspection (see 4.4).
(c) Quality conformance inspection (see 4.5).
(d) Inspection comparison (see 4.7).
(e) Inspection of packaging (see 4.8).

4.3 Preproduction inspection.

4.3.1 Examination. The preproduction section shall be examined as specified in 4.6.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The preproduction section shall be tested as specified in 4.6.2. Failure of any test shall be cause for rejection.

4.4 Initial production inspection. When specified (see 3.3), one or more initial production sections will be selected at random by the Government from the sections being produced by production tooling and will be examined as specified in 4.6.1 and tested as specified in 4.6.2 to determine conformance to the requirements of this specification. The inspection will be performed by the Government at a site selected by the Government. Acceptance of an initial production section shall not exclude the remaining sections from the quality conformance inspection and acceptance provisions specified in Section 4. In addition to any test specified as part of the initial production test, the Government reserves the right to conduct any or all other tests contained in this specification as part of the initial



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production test, and failure of such additional tests shall have the same effect as failure of those tests specified as initial production tests.

4.4.1 Inspection failure. Failure of an initial production section to meet any requirement specified herein during and as a result of the examination and tests specified in 4.4 shall be cause for rejection of the initial production section and shall be cause for refusal by the Government to continue acceptance of production sections until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on sections previously accepted and produced under the contract. Any deficiencies found as a result of the initial production inspection will be considered prima facie evidence that all sections accepted prior to the completion of initial production inspection are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.5 Quality conformance inspection.

4.5.1 Examination. Each section shall be examined as specified in 4.6.1. Presence of one or more defects shall be cause for rejection.

4.5.2 Tests. Each section shall be tested as specified in 4.6.2.3 (a and b) and 4.6.2.10. Failure of any test shall be cause for rejection.

4.6 Inspection procedure.

4.6.1 Examination. The sections shall be examined, as specified herein, for the following defects:

101. Material not as specified.
102. Damaged or missing components.
103. Equipment improperly mounted or incompletely assembled.
104. Recessed folding steps not as specified.
105. Electrical equipment not as specified.
106. Interchangeability not as specified.
107. Safety not as specified.
108. Maintenance not as specified.
109. Identification missing, imcomplete, or not legible.
110. Workmanship not as specified.
111. Electromagnetic interference suppression components and network not identical to the approved model, when applicable.



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4.6.2 Tests.

4.6.2.1 Test conditions. Unless otherwise specified in a test, tests shall be performed in sequence listed herein, with equipment from applicable Set List installed. The section shall be without shelter and at the climatic conditions existing at the place of test. The section shall operate as specified herein without maintenance other than the contractor's recommended normal maintenance schedule prepared and submitted by the contractor prior to test. When applicable, the Government-loaned property specified in 3.13 shall be used for the tests. Any test failure attributable to Government-loaned equipment shall not constitute failure of any test specified herein.

4.6.2.2 Low and high temperature storage. The section shall be placed in an environmental test chamber. All components shall be secured in the storage or transport position. The internal chamber temperature shall be lowered to not less than -34° C (-30° F) and maintained at that temperature for not less than 1 hour. Then the chamber temperature shall be raised to 71° C ( 160° F) and maintained for not less than 5 hours at that temperature. After the chamber temperature has returned to ambient conditions, the section shall be examined and tested in accordance tieh 4.6.2.10. Non-conformance to 3.11.1 and the failure criteria of 4.6.2.10 shall constitute failure of this test.

4.6.2.3 Mobility. The sections shall be subjected to the mobility tests as specified herein. All components shall be secured in the storage or transport position. During all mobility tests, all service lights shall be illuminated, turn signals operated on turns, and hazard warning systems actuated on stops, except as otherwise specified herein.

(a) Highway. Tow the section over hard surface highways at subtained speeds between 45 and 55 mph for not less than 2 hours.
(b) Secondary and unimproved roads. Tow the section over secondary roads (gravel and crushed stone) and unimproved roads (dirt and partially surfaced) at speeds between 30 and 40 mph for not less than 4 hours. Braking shall include not less than 15 abrupt stops from a speed of 25 mph in forward gear and not less than 5 abrupt stops from a speed of 5 mph in reverse gear. Turning shall include not less than 20 sharp turns (10 to the right and 10 to the left) at maximum safe speeds in forward gear.
(c) Off-road. Tow the section over off-road terrain at speeds between 10 and 20 mph for not less than 2 hours.
(d) Chatter course. Tow the section over a chatter course until not less than 26,000 either upward or downward accelerations of magnitudes specified herein have been applied to a rear wheel center. The course shall be of such construction that the acceleration magnitudes indicated below can be obtained by adjusting the vehicle speed without the frequency of the



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accelerations exceeding 15 cycles per second (cps) or becoming less than 3 cps. The maximum acceleration magnitude indicated below shall not be exceeded, and accelerations having magnitudes less than the minimum shall be disregarded. The course shall be of such length that a wheel center receives in one pass over the course at least 50 consecutive vertical acceleration cycles of any magnitude less than the maximum specified herein. The accelerations shall be measured by a trail test to establish the required vehicle test speed. The recording instruments shall then be disconnected and the test completed at established speed. The acceleration values used shall be those of the wheel center receiving the highest accelerations. Means for assuring that the vehicle test speed is held constant throughout the test shall be provided. Accelerations shall be measured on a running-gear member, usually an axle or spindle, that has a motion identical to that of the wheel center. The magnitude of the accelerations shall be not more than 6 g nor less than 3 g, where g represents the weight imposed.

(e) Diagonal V-ditches. Tow the section through V-ditches until 60 reversal torsional frame stresses have been applied. The ditches shall be approximately 18 inches deep, 36 inches wide at the top. With the V-ditches at an angle to the direction of travel, drive the vehicle to obtain one rear wheel and one front wheel, of opposite sides, in the ditches at a time. During the test, measure and record suspension articulation.

(f) Mud course. Tow the section through mud not less then 12 inches deep for a distance equal to 3 times the vehicle's overall length. Immobilize the section and retrieve by towing out.

(g) Braking. Demonstrate and record the section braking capability required of the section. In accordance with DOT FMVSS and FMCSR and the California Vehicle Code procedures and requirements, conduct service, emergency, and parking ability demonstrations.

(h) Semitrailer mobility. With the ISO size van bodies semitrailer mounted, perform the following:

(1) Tow the semitrailer in a pivot-turn about the center of the rear running gear for 360 degrees to the left and 360 degrees to the right on a level dry surface.

(2) Measure and record the deviation of off-tracking of the semitrailer with respect to the truck-tractor and compare with DOT FMCSR.

(3) Jackknife the semitrailer at angles of not less than 90 degrees to the left and to the right from straight ahead.

(4) Demonstrate compatibility of the semitrailer to couple and uncouple from commercial truck-tractors and demonstrate the ability of the semitrailer front supports to raise and lower and withstand loaded [illegible] during [illegible]

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(5) Test brake system in accordance with TTMA RP No. 12 and RP No. 44.

At the completion of each phase of this test, the section shall be examined and tested in accordance with 4.6.2.10. Any evidence of damage or failure of the equipment to operate as specified in 3.11.2, and nonconformance to the failure criteria of 4.6.2.10 shall constitute failure of this test.

4.6.2.4 Rail impact. The complete van body, with set installed and mounted on the semitrailer chassis shall be subject to a rail impact test in accordance with MIL-STD-810, Method 516.2, Procedure VI, except there shall be only two impacts (one in each direction) and they shall be at 8 miles per hour. At the completion of this test, the section shall be examined and tested in accordance with 4.6.2.10. Nonconformance to 3.11.3 and to the failure criteria of 4.6.2.10 shall constitute failure of this test.

4.6.2.5 Precipitation. The section shall be placed on a horizontal surface and shall be exposed to natural or simulated rain with not less than 30 mph horizontal wind velocity. The volume of rain impinging on the horizontal surface shall not be less than 1/4 gallon per minute per square foot of wetted horizontal area. Simulated rain shall originate from a source which produces droplets of uniform distribution and velocity, similar to natural rain at the point of contact with the van body. The test shall be conducted as follows:

(a) Expose the top and left side to the rain as described above for a period of 5 minutes.
(b) Repeat (a) with the top and front exposed.
(c) Repeat (a) with the top and right side exposed.
(d) Repeat (a) with top and rear exposed.

Evidence of leakage into the interior of the van body or nonconformance to 3.11.4 and to the failure criteria of 4.6.2.10 shall constitute failure of this test.

4.6.2.6 Leveling. The sections shall be parked on a side slope (perpendicular to the long axis) of approximately 5 percent. With all doors closed, the vehicle shall be leveled by means of the leveling jacks as determined by the van body levels. Check the actual leveling of the van body interior floor. Failure of the doors to open and close without binding or nonconformance to 3.11.5 shall constitute failure of this test.

4.6.2.7 Blackout. With the sections in darkness, close the doors and the door blackout curtains, and turn on all interior lights within the van body. Open and close the personnel doors. Leakage of light or nonconformance to 3.11.6 shall constitute failure of this test.



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4.6.2.8 Demounting. The sections shall have all items secured within the van body, all underbody appurtenances secured, and the doors closed. Unlatch the van body from the chassis, lift the van body, and place on the ground. Then lift the van body, reposition it back on the chassis and latch down. Inability to remove and reinstall the van body on the chassis, malfunction, permanent deformation, or damage to the van body or vehicle, or nonconformance to 3.11.7 shall constitute failure of this test.

4.6.2.9 Electromagnetic interference. The sections shall be tested to determine conformance to 3.11.8. The contractor shall furnish the contracting officer a report of tests. Disapproval of the report shall constitute failure of this test.

4.6.2.10 Operational. The operational test shall be conducted at each time as specified herein. Examine the section for damage, connect the section to the generator, operate all equipment include in the sections including air conditioners, heaters, and lights. Any malfunction, breakage, damage, displacement, defective welded, bolted, or riveted connections, inability to operate or permanent deformation of the section equipment shall constitute failure of this test.

4.6.2.11 Human Engineering Demonstration. The contractor shall demonstrate adherence to the criteria established in paragraph 3.7. Nonconformance to 3.7 shall constitute failure of this demonstration.

4.6.2.12 Maintainability. When specified (see 6.2) the requirements of 3.10 shall be demonstrated by analysis of all data generated during the conduct of the tests and inspections required during the procurement. A formal demonstration/analysis shall be performed, in accordance with government approved test plan.

4.6.2.13 Preparation for operation and movement. This test shall begin with the section prepared for transport. The section shall be completely set up and prepared for field operation within two (2) hours. This shall include the unpacking, unsecuring, demounting, set up and preparing as required of all equipment and materials; and the leveling, connecting power cables and all other preparations required for field operation of the van body. After this is completed, the section shall be packed, secured and prepared for transport within two (2) hours. This shall include the packing, storing, securing demounting and cleanup as required of all equipment and materials; all water tanks shall be emptied; chemicals, all leveling jacks, power cables, waterhoses shall be properly stored; and all other required preparations of the van body prior to transport. The number of personnel used for this test shall not exceed the number assigned to operate the section. Nonconformance to 3.11.9 shall constitute failure of this test.



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4.7 Inspection comparison. The Government may select sections at any time during the contract production period and subject these sections to the examination and tests specified in 4.4 to determine conformance to the requirements of this specification. The inspection will be performed by the Government, at a site selected by the Government, on sections selected at random from those which have been accepted by the Government and will not include the previously inspected preproduction model and initial production sections. In addition to any test specified as part of the inspection comparison, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison, and failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparison.

4.7.1 Inspection failure. Failure of an inspection comparison section to meet any requirement specified herein during and as a result of the examination and tests specified in 4.7 shall be cause for rejection of the inspection comparison section and shall be cause for refusal by the Government to continue acceptance of production modules until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on sections previously accepted and producted under the contract. Any deficiencies found as a result of the inspection comparison will be considered prima facie evidence that all sections accepted prior to the completion of inspection comparison are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.8 Inspection of packaging.

4.8.1 Quality conformance inspection of pack.

4.8.1.1 Unit of product. For the purpose of inspection, a section prepared for shipment shall be considered a unit of product.

4.8.1.2 Examination. Each section shall be examined for the following defects:

| | |
|---|---|
| 112. | Materials, methods, and containers not as specified for Level A. Each incorrect material, method, or container shall be considered one defect. |
| 113. | Unprotected exterior surfaces not coated with preservatives as specified for Level A. |
| 114. | Coated machine surfaces not wrapped or covered with barrier material as specified for Level A. |
| 115. | Highly critical components not preserved as specified for Level A. |



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116. Items not cushioned as specified for Level A.
117. Sets or kits not contained in the carrying case when carrying case is furnished as specified for Level A.
118. Mounting brackets and holders not utilized as specified for Level A.
119. Consolidation not as specified for Level A.
120. Semitrailer components not preserved in accordance with the referenced document as specified for Level A.
121. Openings not sealed as specified for Level A.
122. Packing not as specified.
123. Marking missing, illegible, incorrect, or incomplete.

## 5. PACKAGING

5.1 Preservation. Preservation shall be Level A or Commercial, as specified (see 6.2).

5.1.1 Level A. Components shall be provided presevation adequate to prevent deterioration and damage against the most severe conditions known or anticipated to be encountered during shipment, multiple handling, and indeterminate open storage. The preservation requirements for specific components to meet the above requirements shall be determined by the contractor, utilizing the criteria established in MIL-P-116. The following minimum requirements shall be adhered to.

5.1.1.1 Preservatives. Preservatives specified herein shall conform to the applicable specification listed in and shall be applied in accordance with MIL-P-116.

5.1.1.2 Unprotected surfaces. Unprotected exterior metal surfaces of components requiring the application of a contact preservative in accordance with MIL-P-116 shall be preserved as follows.

5.1.1.2.1 Unfinshed surfaces. Unfinished surfaces shall be coated with Type P-1 preservative.

5.1.1.2.2 Machined surfaces. Exposed exterior machined surfaces shall be coated with Type P-6 or P-11 preservatives. The coated surfaces shall be wrapped or covered with barrier material conforming to MIL-B-121, Type I, Grade A, Class 2 and secured in place with tape conforming to PPP-T-75, Type III.

5.1.1.3 Highly criticial components. Mechanical or electrical components of a highly critical nature which require the highest degree of protection from the effects of water vapor shall be preserved in accordance with MIL-P-116, Method II, submethod as applicable.



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5.1.1.4 Cushioning. Items requiring protection from physical or mechanical damage shall be protected by wrapping, cushioning, pack compartmentization,cartonizing, or other means to mitigate shock and vibration.

5.1.1.5 Sets or kits. When items are furnished in sets or kits with a carrying case, the items shall be preserved as specified herein and when possible, replaced in the case.

5.1.1.6 Components having mounting brackets or holders. Components furnished with mounting brackets or holders shall be preserved as specified herein and, when possible, replaced in the bracket or holder and secured to prevent movement and damage.

5.1.1.7 Consolidation. Components that cannot be placed in compartments or drawers or secured by the mounting brackets or holders provided, shall be consolidated together in close-fitting boxes conforming to PPP-B-601, Domestic Type, style optional. The boxes shall be of a size to fit into the most convenient space for securing. Blocking, bracing, cushioning, and anchoring shall be in accordance with MIL-STD-1186. All doors and drawers shall be secured to prevent opening during shipment and handling.

5.1.1.8 Semitrailer. Semitrailer components shall be preserved in accordance with the Level A (mobile) requirements of MIL-STD-281.

5.1.1.9 Openings. Openings to the exterior of the ISO containers from the air conditioner or heating system shall be sealed with tape conforming to PPP-T-75, Type II or a combination of the tape and paper conforming to PPP-B-1055, Class E-1 or E-2.

5.1.2 Commercial. Preservation shall be in accordance with MIL-STD-1188 and all components secured in a manner to prevent movement or damage during shipment and handling.

5.2 Packing (see 6.8). Each section, preserved as specified in 5.1, shall be prepared for shipment in a manner to assure carrier acceptance and safe delivery to destination at lowest ratings in compliance with Uniform Freight Classification rules or National Motor Freight Classification rules. Doors shall be secured to prevent accidental opening and in a manner to prevent pilferage of contents.

5.3 Marking.

5.3.1 Military. Marking for military levels of protection shall be in accordance with MIL-STD-129.



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5.3.2 Commercial. Marking for commercial packaging shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The intended use of the Topographic Support System (TSS) is a combat support unit consisting of functional sections (within subsystems). TSS will provide the field commander with complete topographic support in a response time commensurate with tactical requirements. Each subsystem will consist of one or more sections (modules), each dedicated to one or more phases of subsystem operation. The sections will be housed in a transportable container (van body) modified for human habitation and equipped with the necessary instrumentation to perform the required topographic support operations.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Title, number, and date of applicable specification sheets (see 1.2 and Supplement).
(c) Time frame required for submission of preproduction model (see 3.2).
(d) When the Government will conduct any or all of the preproduction model examination and tests. When the Government will conduct some but not all of the preproduction examination and tests, the contracting officer should specify which examination and tests will be conducted by the Government and which examination and tests shall be conducted by the contractor (see 3.2).
(e) When initial production inspection is required, and number of sections required (see 3.3).
(f) When rail impact test is required (see 3.11.3).
(g) When electromagnetic suppression is required (see 3.11.8).
(h) When Government-loaned property is required (see 3.13).
(i) When a formal maintainability demonstration analysis is required (see 3.10 and 4.6.2.12).
(j) Degree of preservation required (see 5.1)

6.3 Preproduction model. Any changes or deviations of production sections from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the contractor of his obligation to furnish sections conforming to this specification.

6.4 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of this specification (see 3.4).



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6.5 Government property. The contracting officer should arrange to lend the property specified in 3.13. The contracting offices should arrange to furnish the air conditioners designated in the set listings.

6.6 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and contractor's maintenance and operation manual to be furnished with each section.

6.7 Provisioning. The contracting officer sould include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools), and instructions regarding shipment of sections.

6.8 Packing. Degree of packing has not been included, as the packing specified is acceptable for any degree of protection.

6.9 Definition. The following definitions shall apply throughout this specification:

Section and module are synonymous.
Van body and ISO container are synonymous.
Chassis and semitrailer are synonymous.
Set listing and component list are synonymous.
Damage shall include malfunction, breakage, inability to operate or permanent deformation.

6.10 Topographic Support System (TSS). The total system is composed of the following structures:

Command and Control Subsystem
  Operation Section
Storage, Retrieval and Distribution Subsystem
  Storage and Retrieval Section
  Distribution Section
Cartographic Revision Subsystem
  Compilation Section
  Drafting Support Section
Survey Subsystem
  Survey Section
Image Base Product Subsystem
  Rectifier I Section
  Rectifier II Section
  Mosaicking/Drafting Section
Military Geographic Information Subsystem
  Collection Section
  Analysis Section
  Information Section



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Synthesis Section
Direct Support Section
Reproduction Subsystem
Camera Section
Press Section
Map Layout Section
Plate Process Section
Photo Mechanical Process Section
Finishing Section
Paper Conditioning Section

The Reproduction Subsystem (FSC 3610) is under specification MIL-T-52966(ME)

The Direct Support Section (FSC 6675) is under specification MIL-T-52965(ME).

Custodian:
Army - ME

Preparing activity:
Army - ME

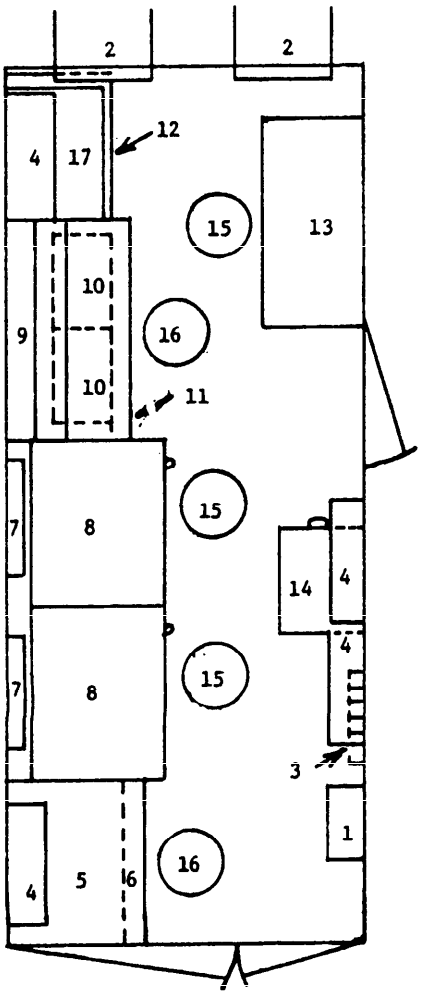
Review activity:
Army - TE

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SC 6675-97-CL-E54




| No. | Item |
|---|---|
| 1 | Panel Box (Contractor Installed) |
| 2 | Air Conditioner |
| 3 | Kit, Mounting Rifle |
| 4 | Cabinet, Storage, Wall |
| 5 | Filing Cabinet, Map & Plan (Flat) |
| 6 | Drawing Board |
| 7 | Board Tracing, Portable |
| 8 | Desk, Flat Top |
| 9 | Copying Machine, Diazo |
| 10 | Cabinet, Storage |
| 11 | Table, Office |
| 12 | Table, Office |
| 13 | Table, Light: Split Stage |
| 14 | Filing Cabinet, Safe |
| 15 | Chair, Rotary, Desk |
| 16 | Chair, Rotary, Drafting |
| 17 | Facsimilie Receiver |

Equipment as shown in approximate location only and the exact location will be accomplished by the Contractor in conjunction with the Government after the factors of human engineering and safety have been approved.

"NOT TO SCALE"

FIGURE 1. Direct support section (TSS), truck mounted

X-3445A



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE MIL-T-52964(ME) Topographic Support System (TSS): Operation, Storage & Retrieval, Dftg Supt, Compilation, etc, Gen Spec

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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